Expression of Interest – Deloitte Capstone Program

Team Name: InsightForge
Team Captain: Eshwar B N
Members: Harsha Madev Hegde, Ruchitha M

1. Introduction

We are PES University engineering students with Computer Science specialisation and have a profound competency in Artificial Intelligence, Full Stack Engineering and System Design. As part of our Capstone Project, we are developing solutions to real life examples of urban mobility by providing an Intelligent Solution (IS) that combines multiple modes of transport. This is why we want to be involved with the Deloitte Capstone program, to leverage business mentors and gain exposure to practical experiences that will improve our solution.

2. Problem Statement

Urban commuters face significant challenges due to fragmented transportation systems. Planning a journey requires switching between multiple platforms for buses, metro, ride-hailing services, and private vehicles. This leads to inefficiency, increased costs, and poor decision-making.

Our project addresses this problem by eliminating the need for multiple applications and providing a unified, intelligent system for comparing and optimizing travel options.

3. Proposed Solution

We propose a **Smart Urban Mobility Aggregator**, a unified platform that integrates:

- Public transport (BMTC buses and BMRCL metro)
- Ride-hailing services (Ola, Uber, Rapido, Namma Yatri)
- Personal vehicle cost estimation

The system enables users to:

- Compare travel options based on **cost, time, and convenience**
- Receive **AI-driven recommendations** tailored to user preferences
- View optimized routes using graph-based algorithms
- Access personalized travel suggestions based on usage patterns

Our system leverages **Dijkstra's Algorithm for routing**, structured datasets (CSV, GTFS), and API-based real-time data integration to deliver accurate and efficient results.

4. Technologies Used

- **Backend:** Python (routing, data processing, optimization)
- **Frontend:** Flutter (mobile), React (web)
- **Database:** Firebase (cloud-based storage)
- **Algorithms:** Graph-based routing (Dijkstra), recommendation scoring
- **APIs:** Google Maps APIs for distance and travel estimation

5. Expected Impact

- Simplifies urban commuting through a **single unified interface**
- Enables **cost-efficient and time-optimized travel decisions**
- Reduces dependency on multiple applications
- Provides a scalable solution for **smart city ecosystems**

6. Innovation & Differentiation

Unlike existing solutions that focus on a single mode of transport, our platform integrates **multi-modal transportation** into one system. It combines:

- Graph-based optimization for public transport
- Fare comparison across ride-hailing platforms
- Fuel-based cost estimation for private vehicles
- Personalized recommendations based on user preferences

This makes the system not just a comparison tool, but an **intelligent decision-making assistant for urban mobility**.

7. Why Our Team

Our team has successfully designed and implemented independent modules for each transportation mode and validated them for accuracy and performance. We bring strong expertise in AI, data processing, and full-stack development, along with a clear vision to build scalable and impactful systems.

8. Conclusion

We look forward to leveraging Deloitte's mentorship to enhance our system with advanced AI capabilities, real-time data integration, and business scalability. This program will enable us to transform our solution into a practical, industry-ready smart mobility platform.